

Comparative Bioequivalence of Single-Dose Dapagliflozin, Glimepiride, and Extended-Release Metformin Tablets in Healthy Adult Volunteers: Findings from a Randomized, Two-Way Crossover, Open-Label Study

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INTRODUCTION

- Type 2 diabetes mellitus (T2DM) is a metabolic disorder and a leading cause of morbidity and mortality.¹
- Achieving and maintaining tight glycaemic control is an essential component in managing T2DM and may necessitate the use of various antidiabetic medications.²
- If adequate blood glucose control is not attained using monotherapy, a combination of agents with different mechanisms of action may result in better glycaemic control.³
- Both Dapagliflozin and Glimepiride significantly improve glycaemic control, while Metformin improves insulin sensitivity.^{4,5}
- Dapagliflozin, Glimepiride and Metformin Hydrochloride has complementary mechanisms of action, and its fixed dosage combination (FDC) may offer simpler treatment regimen for T2DM patients, lowering pill burden and improving compliance and outcomes.⁶

OBJECTIVE

- To assess comparative bioequivalence of FDC that is a bilayer tablet containing Dapagliflozin, Glimepiride, and Metformin (10mg / 2mg / 1000mg) (by SunPharma Ltd.) in comparison to reference products [Xigduo XR (Dapagliflozin 10 mg/ ER Metformin HCl 1000mg Tablets; by AstraZeneca Pharma India Ltd.) and Amaryl Tablets (Glimepiride 2 mg; by Sanofi India Ltd.)] administered separately in healthy adult human subjects under fed condition.

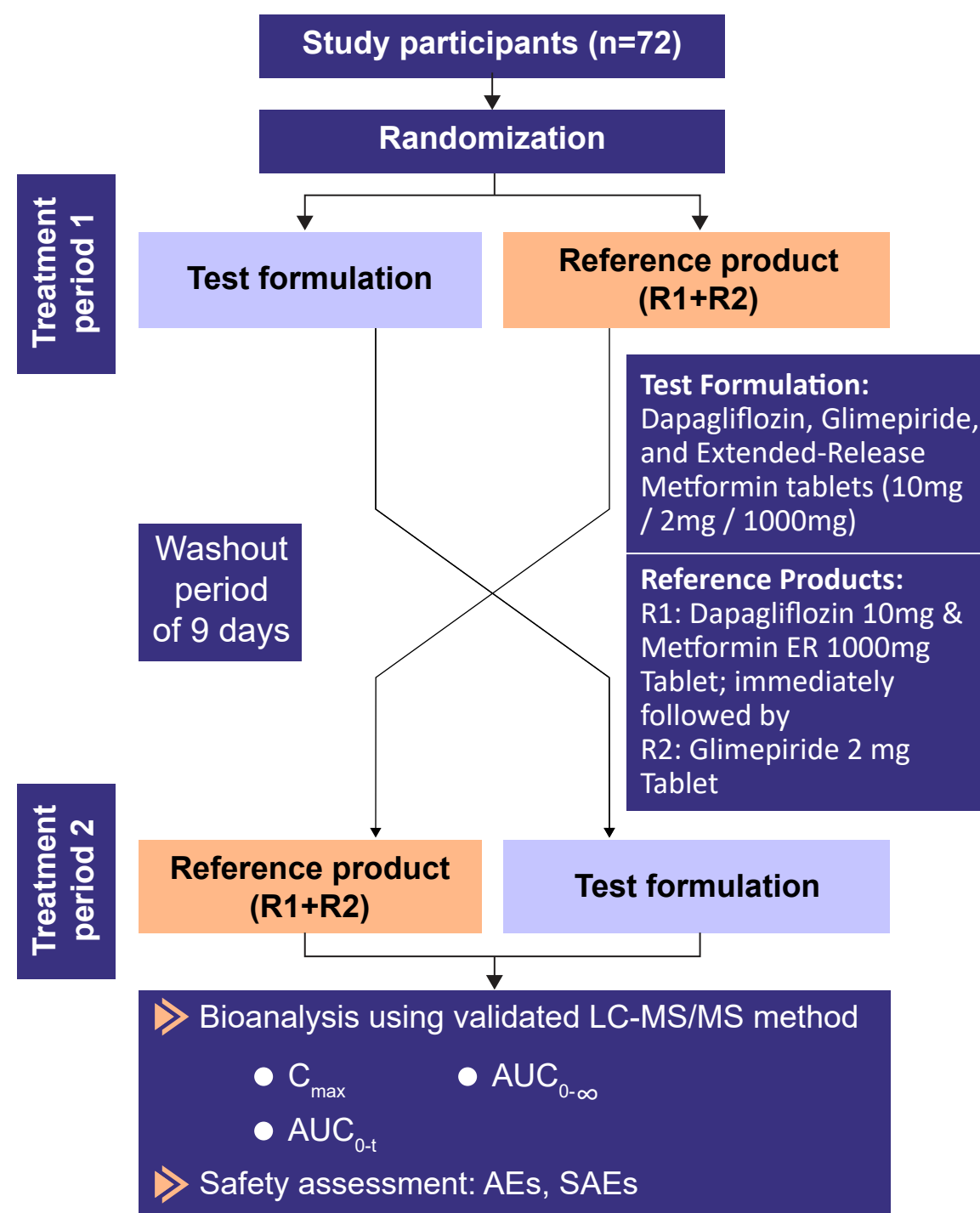
METHODOLOGY

- **Study design:** An open label balanced, randomized, two treatment, two sequence, two period, single dose, crossover bioequivalence study in healthy adult human subjects under fed condition
- **Inclusion criteria:** Adult healthy volunteers aged 18-50 years with body mass index (BMI) ≥ 18.50 and < 30.00 kg/m², haemoglobin level ≥ 13.0 g/dL for males or ≥ 12.0 g/dL for females and who voluntarily gave written informed consent to participate in this study

➤ Exclusion criteria:

- Subjects with history of hypersensitivity, recurrent impaired renal function or renal disease or dialysis, any infection, fever, vomiting and/ or diarrhoea, malaise, myalgia, respiratory distress or abdominal pain, drug induced rash and/or pruritus, serious medical illnesses, drug dependence or excessive alcohol intake.
- Subjects having evidence of organ dysfunction or any clinically significant abnormality, had used any medication within 30 days, had donated blood or received an investigational medicinal product, or consumed alcohol, grapefruit juice and or grape fruit supplements 48 hours prior to admission.
- Subjects who were regular smoker, had inability to communicate well, or were pregnant or breast feeding females.

Figure 1: Study flow chart from screening to study completion



RESULTS

- Of the 72 subjects enrolled, 63 subjects completed the study.

Table 1: Pharmacokinetic parameters following administration of Test and Reference products

Drug	Parameter	Least Squares Geometric Means		% Ratio (T/R)	90% CI
		Test product	Reference product		
Dapagliflozin	AUC0-t	661.35	691.62	95.62	93.88- 97.40
	AUC0-∞	682.65	712.64	95.79	94.17- 97.44
	Cmax	119.63	122.58	97.60	91.53-104.07
Glimepiride	AUC0-t	1168.39	1206.22	96.86	93.55-100.29
	AUC0-∞	1200.66	1236.31	97.12	93.85-100.50
	Cmax	195.88	183.47	106.76	100.93-112.93
Metformin	AUC0-t	13538.52	97.60	97.60	94.10- 101.23
	AUC0-∞	13812.35	97.32	97.32	93.96-100.80
	Cmax	1059.63	93.18	93.18	88.60-97.99

*Arithmetic mean presented; #Median (Min-Max) presented

SAFETY ASSESSMENT

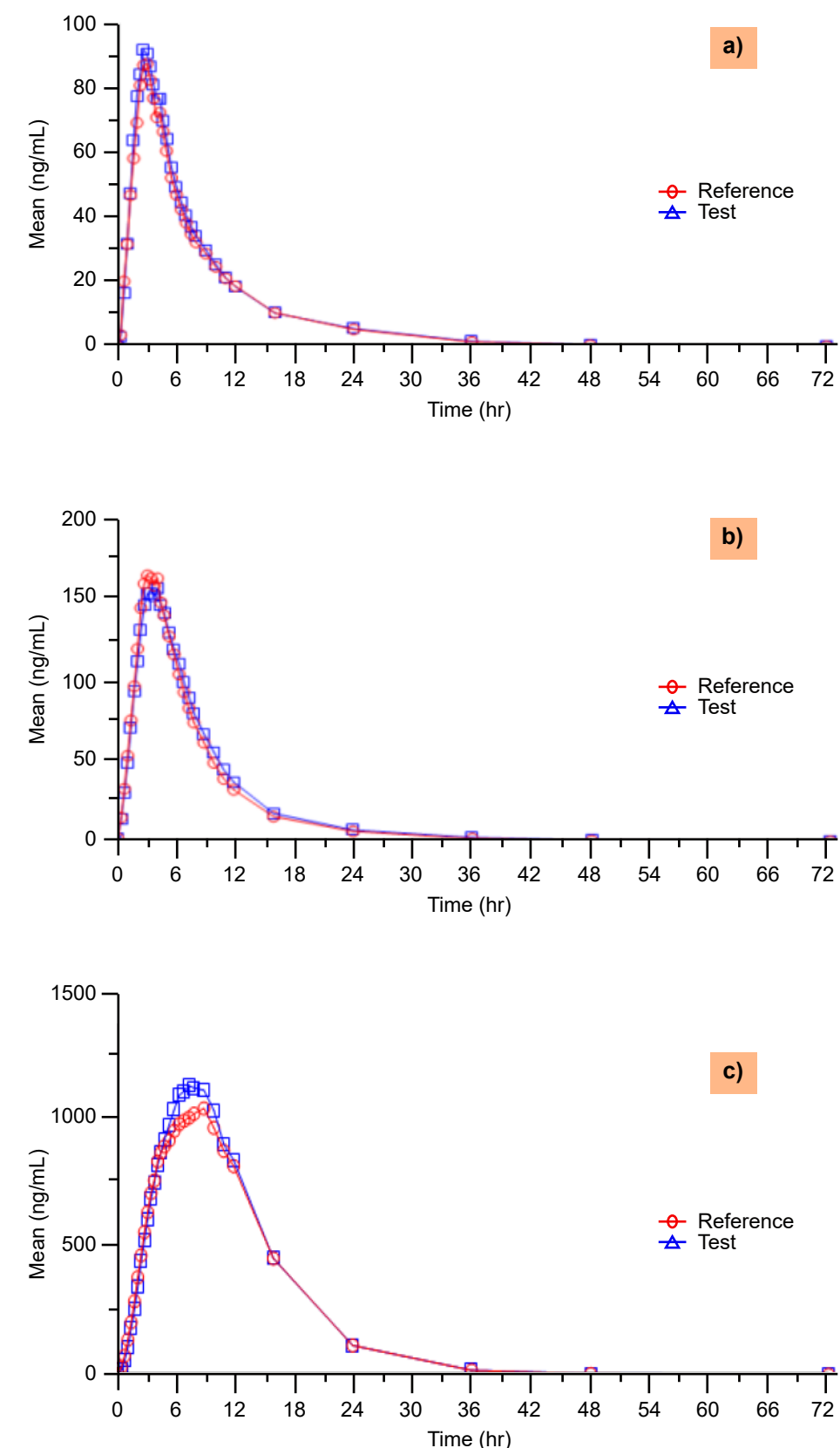
- A total of five AEs were recorded, three with Test Product and two AEs with Reference Product.
- All AEs were mild to moderate in severity, which resolved without sequelae.
- No instances of SAEs were documented during the study.

References:

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5. Herman, Rok, et al. "Metformin and insulin resistance: A review of the underlying mechanisms behind changes in GLUT4-mediated glucose transport." *International journal of molecular sciences* 23.3 (2022): 1264.
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Abbreviations: AEs: Adverse events; AUC: Area under the receiver operating characteristic (ROC) Curve; Cmax: Maximum concentration; FDC: Fixed dose combination; LC-MS: Liquid chromatography–mass spectrometry; LSM: Least-squares mean; SAEs: Serious adverse events; T2DM: Type 2 diabetes mellitus.

Figure 2: Mean Plasma Concentration - Time profile of A) Dapagliflozin B) Glimepiride and C) Metformin in Test and Reference products



DISCUSSION AND CONCLUSION

- Single oral dose of FDC tablet of Dapagliflozin, Metformin, and Glimepiride (10 mg/ 2 mg/ 1000 mg) is bioequivalent to individual reference products in healthy adults under fed conditions, with pharmacokinetic parameters (Cmax, AUC0-t and AUC0-∞) within the 80-125% bioequivalence range.
- Both test and reference products were well tolerated by the study subjects. There were no SAEs reported in the study.
- The study supports the FDC as a viable, effective, and well-tolerated treatment option in diabetes care, especially for patients who require multiple antidiabetic agents.